



Islamic University of Technology
Department of Computer Science and Engineering

Lab 4: Shortest Path Algorithm

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

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Task A. Mandatory Stopover

Time Limit: 2 seconds | Memory Limit: 256 MB

A courier must deliver an urgent package from city S to city D . However, the package is not yet ready — the courier must first travel to a processing depot at city C to collect it, and only then proceed to the destination.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of x_i . The courier always takes the fastest available route.

Find the minimum total travel time for the courier to go from city S to city D with a mandatory stop at city C .

Input Format

The first line contains five integers N, M, S, D and C ($2 \leq N \leq 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D, C \leq N, S \neq D$).

Each of the next M lines contains three integers u_i, v_i and x_i ($1 \leq u_i, v_i \leq N, u_i \neq v_i, 1 \leq x_i \leq 10^9$) describing a bidirectional road.

It is guaranteed that a valid route always exists. There may be multiple roads between the same pair of cities.

Output Format

Output a single integer — the minimum travel time from city S to city D passing through city C .

Examples

Sample Input	Sample Output
4 4 1 4 3 1 2 1 2 4 1 1 3 5 3 4 1	6
5 6 2 5 3 1 2 2 1 3 4 2 3 1 3 4 3 2 5 8 4 5 2	6

Notes

In the first sample, $S = 1, D = 4, C = 3$. The courier travels $1 \rightarrow 3 \rightarrow 4$ with a total travel time of $5 + 1 = 6$. Although the road $1 \rightarrow 2 \rightarrow 4$ costs only 2, it does not pass through the depot at city 3.

In the second sample, $S = 2, D = 5, C = 3$. The optimal route is $2 \rightarrow 3 \rightarrow 4 \rightarrow 5$ with a total travel time of $1 + 3 + 2 = 6$.

Task B. Road Inspection

Time Limit: 2 seconds | Memory Limit: 256 MB

A road maintenance crew must travel from city 1 to city N along the shortest route. Once they arrive, they need to file a report identifying the single most time-consuming road segment they used — the one with the highest travel time — so it can be prioritised for inspection.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of x_i . Find the maximum travel time among all road segments on the shortest path from city 1 to city N .

Input Format

The first line contains two integers N and M ($2 \leq N \leq 10^5$, $1 \leq M \leq 3 \times 10^5$).

Each of the next M lines contains three integers u_i , v_i and x_i ($1 \leq u_i, v_i \leq N$, $u_i \neq v_i$, $1 \leq x_i \leq 10^9$) describing a bidirectional road.

It is guaranteed that city N is reachable from city 1 and that the shortest path from city 1 to city N is unique. There may be multiple roads between the same pair of cities.

Output Format

Output a single integer — the maximum road travel time on the shortest path from city 1 to city N .

Hints

Track the parent of each city during Dijkstra to reconstruct the shortest path.

Examples

Sample Input	Sample Output
5 5 1 2 3 1 3 1 3 4 2 4 5 4 2 5 10	4
4 4 1 2 2 2 3 1 1 3 6 3 4 3	3

Notes

In the first sample, the shortest path is $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$ with total travel time $1 + 2 + 4 = 7$. The road segments have travel times 1, 2, and 4, so the maximum is 4.

In the second sample, the shortest path is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ with total travel time $2 + 1 + 3 = 6$. The maximum road travel time on this path is 3.

Marks Distribution

Task	Marks
Task A	50%
Task B	50%

Practice Problems

Problem	Links
Shortest Routes I	CSES: https://cses.fi/problemset/task/1671
Jzzhu and Train Routes	Codeforces: https://codeforces.com/problemset/problem/449/B
Dijkstra?	Codeforces: https://codeforces.com/problemset/problem/20/C
Sending Email	UVA: https://onlinejudge.org/index.php?option=onlinejudge&Itemid=8&